

HW-2: Simpler Transceiver Design for PAM-2 and 16-QAM

1. Design a MATLAB program to realize a simple transceiver, as shown in Fig. 1, including a mapper, a channel, a de-mapper and a slicer for PAM-2 and 16-QAM modulation schemes, where the channel model is assumed as an ideal channel, namely, $h_n = 1$. A test bit streams (“b.dat”) can be founded in E-Learning. The total number of bits is 100.

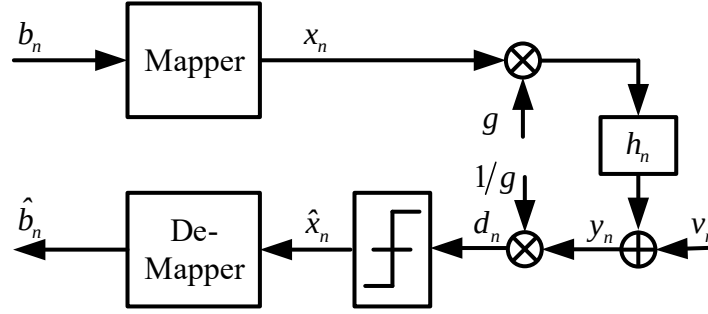


Fig. 1: Simple transceiver including a mapper, a channel, a de-mapper and a slicer.

Actually, the slicer is non-linear device that is applied to realize a series of decision thresholds for the input signal. The related operations of the slicer for PAM-2 and 16-QAM are described as follows,

- (1) For PAM-2, the decision threshold of the slicer is listed as

$$\hat{x}_n = \begin{cases} +1, & y_n \geq 1 \\ -1, & y_n < 0 \end{cases}$$

- (2) For 16-QAM, the decision thresholds of the slicer are 0 and ± 2 that can be easily found in Fig. 2. Considering an example for the real-part of an input signal, the slicer operation is described as

$$\hat{x}_{R,n} = \begin{cases} +3, & y_{R,n} \geq 2 \\ +1, & 0 \leq y_{R,n} < 2 \\ -1, & -2 \leq y_{R,n} < 0 \\ -3, & y_{R,n} < -2 \end{cases}$$

where $\hat{x}_{R,n}$ and $y_{R,n}$ are the real-part for $\hat{x}_n$ and y_n , respectively. Similarly, the imaginary-part of an input signal is the same as the decision thresholds of the real-part of an input signal.

- (3) The gain factors, g and $1/g$, are used to normalize the signal power of x_n to one, where g is defined as follows,

$$g = \begin{cases} 1, & \text{PAM-2} \\ 1/\sqrt{10}, & \text{16-QAM} \end{cases}$$

- (4) The white Gaussian noise with zero mean and unit standard deviation ($\sigma_v \approx 1$) can be modelled as below.

- (a) For PAM-2, the v_n is a real value. A template MATLAB for white Gaussian noise is given as

:

$$v_n = \sigma_v * randn(1);$$

:

- (b) For 16-QAM, the v_n is a complex value. A template MATLAB for white Gaussian noise is given as

:

$$v_n = \sigma_v * \left(\frac{randn(1)}{\sqrt{2}} + j * \frac{randn(1)}{\sqrt{2}} \right);$$

:

- (5) As a matter of fact, $\hat{x}_n = x_n$ and $\hat{b}_n = b_n$ for PAM-2 and 16-QAM

Problem:

Q1. You can use **MTyp** to configure the slice type for PAM-2 and 16-QAM. For example, **MTyp** = 0 for **PAM-2**, and **MTyp** = 1 for **16-QAM**.

Q2. Generate the white Gaussian noise -12 dB for PAM-2 and -20 dB for 16-QAM.

Q3. Evaluate the functions for the simple transceiver. In other words, you need to evaluate $\hat{x}_n = x_n$ and $\hat{b}_n = b_n$.

Q4. Give the input bits of 10^6 and find the BER from SNR = 0 dB to 14 dB for PAM-2, and from SNR = 0 dB to 24 dB for 16-QAM.

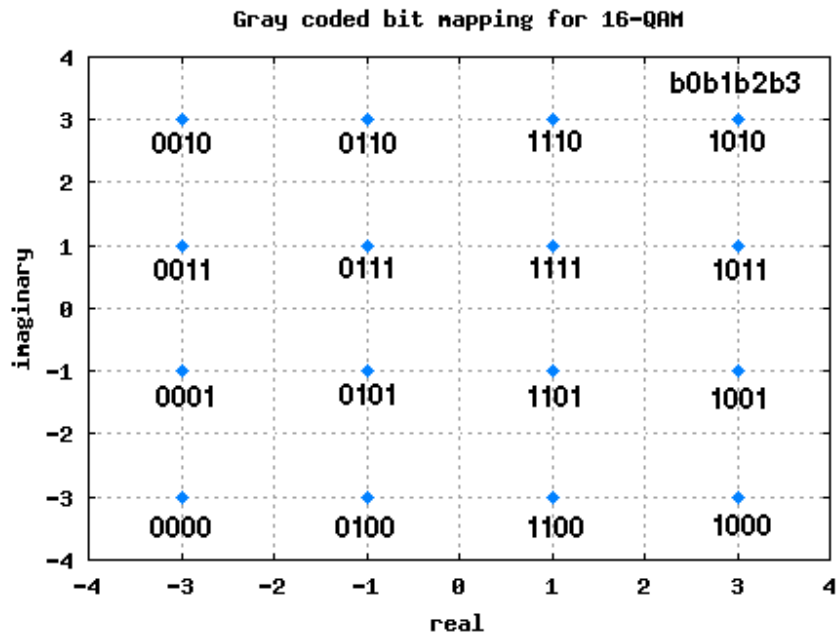


Fig. 2: 16-QAM Constellation